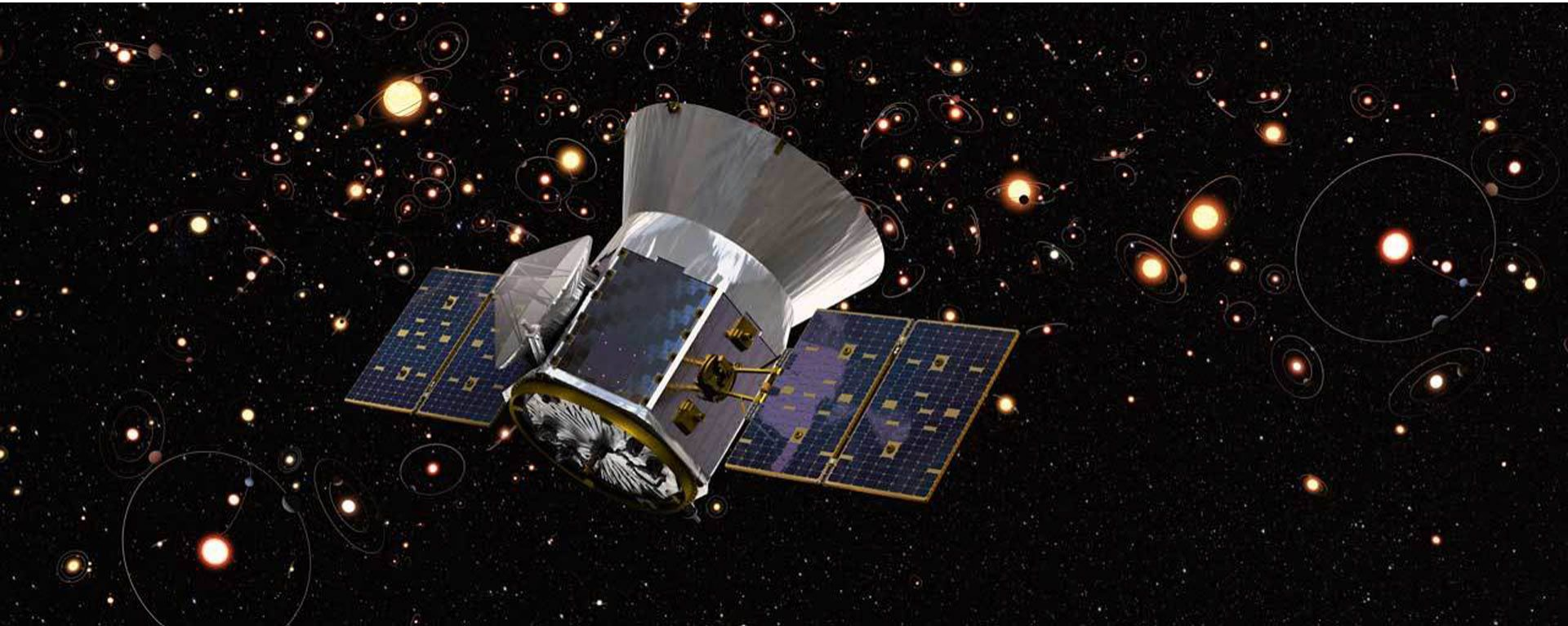


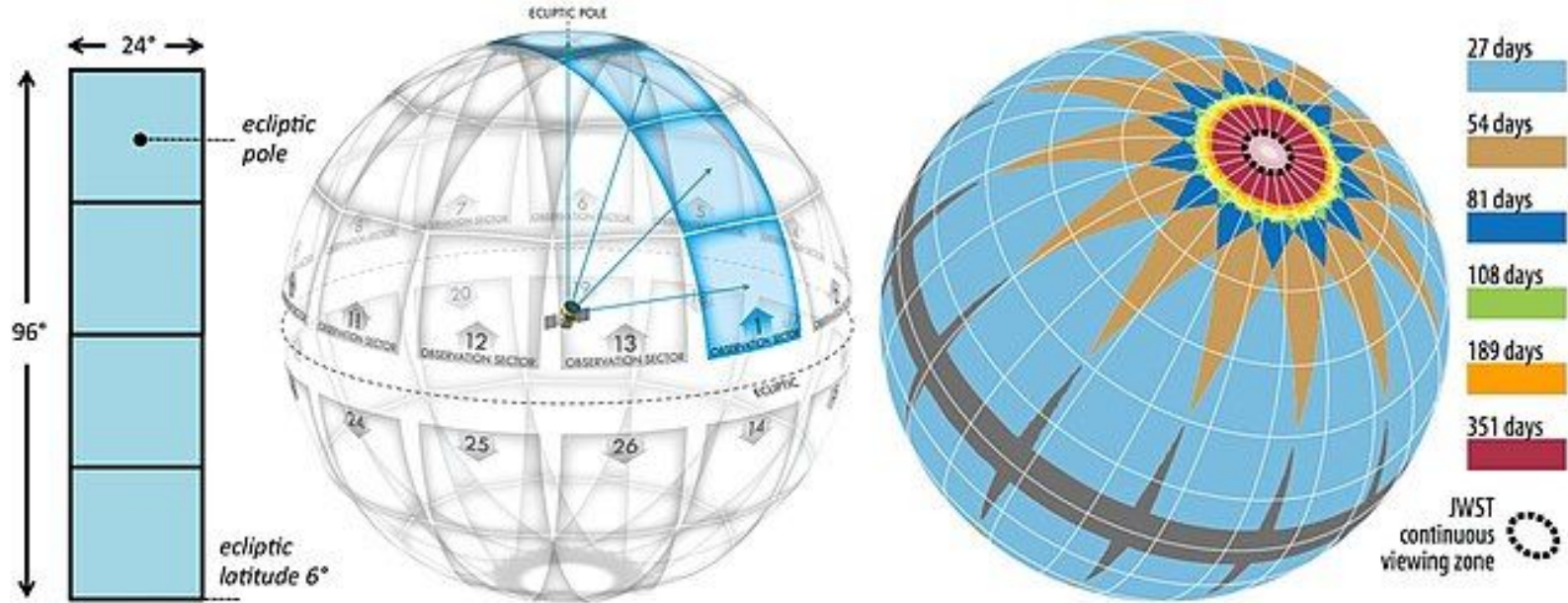
A Signal in the Noise: Using Machine Learning to Improve TESS's Systematic Error and Find More Planets

Arielle Frommer, ASTRON 99, April 4th 2025



Since 2018, the **Transiting Exoplanet Satellite Survey (TESS)** has been searching for exoplanets via the **transit method**

TESS collects continuous light curves for ~30 days at a time in each **sector**, covering the entire sky every 2 years



Source: Ricker et al. 2015

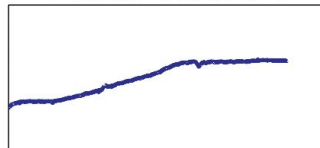
Systematic Noise

- Systematic noise from the telescope can obscure transits
- Sources of noise include pointing drift, focus changes, thermal gradients across the spacecraft, cosmic rays, straylight from the Earth and Moon, etc.
- TESS uses **Pre-search Data Conditioning (PDC)** to remove systematic noise

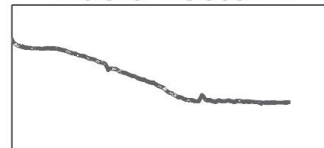
PDC Noise Removal

- Removes systematic noise (ie. telescope artifacts or straylight)
- Preserves unique transit signals and intrinsic stellar variability
- Identifies most common features (the noise) → stores them as vectors → subtracts them from data

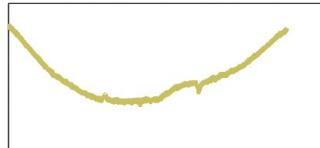
Basis Vector 1



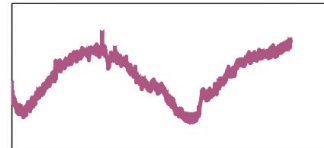
Basis Vector 2



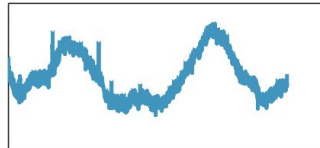
Basis Vector 3



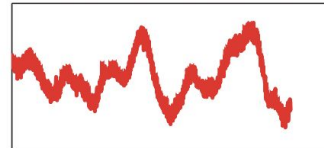
Basis Vector 4



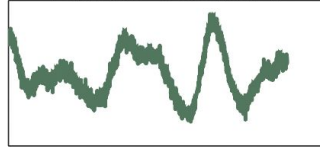
Basis Vector 5



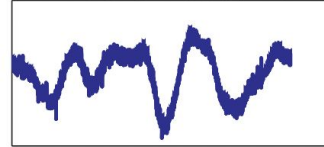
Basis Vector 6



Basis Vector 7



Basis Vector 8

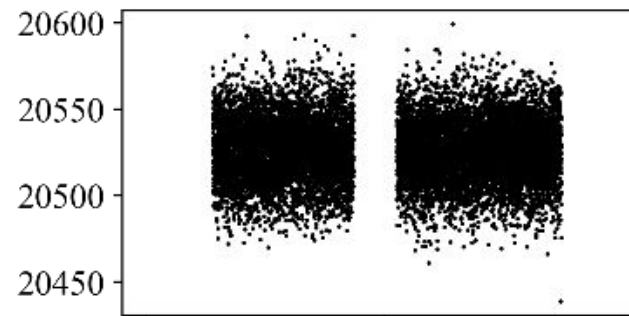
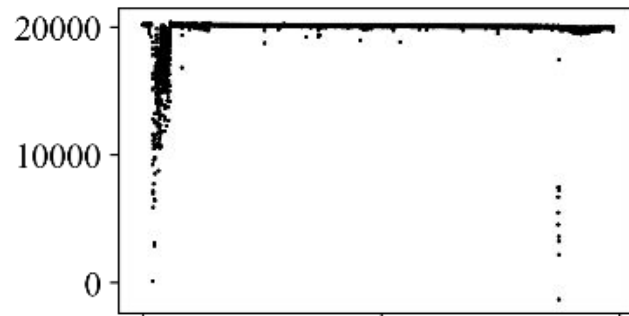


PDC Noise Removal

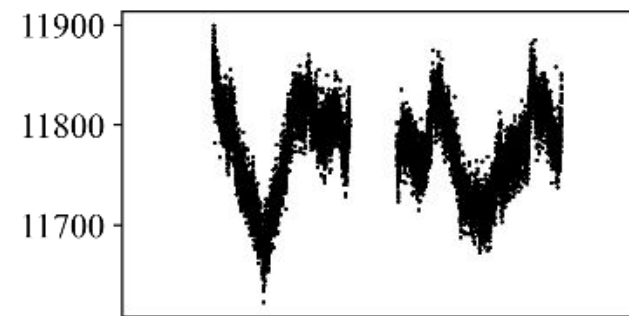
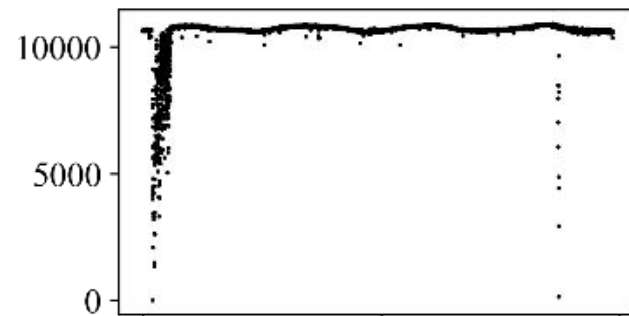
Raw flux

PDC flux

TIC 364588606

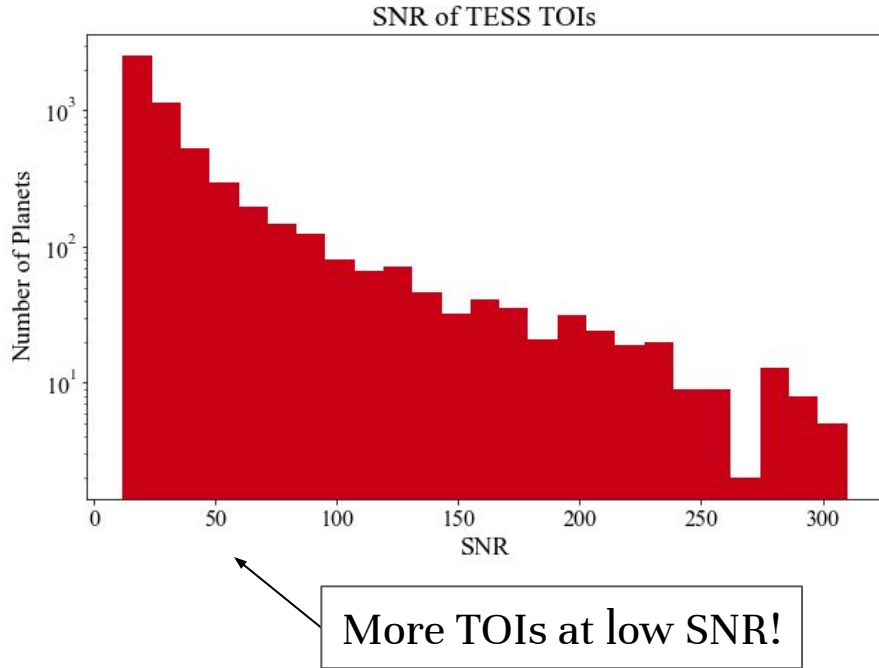


TIC 167164775



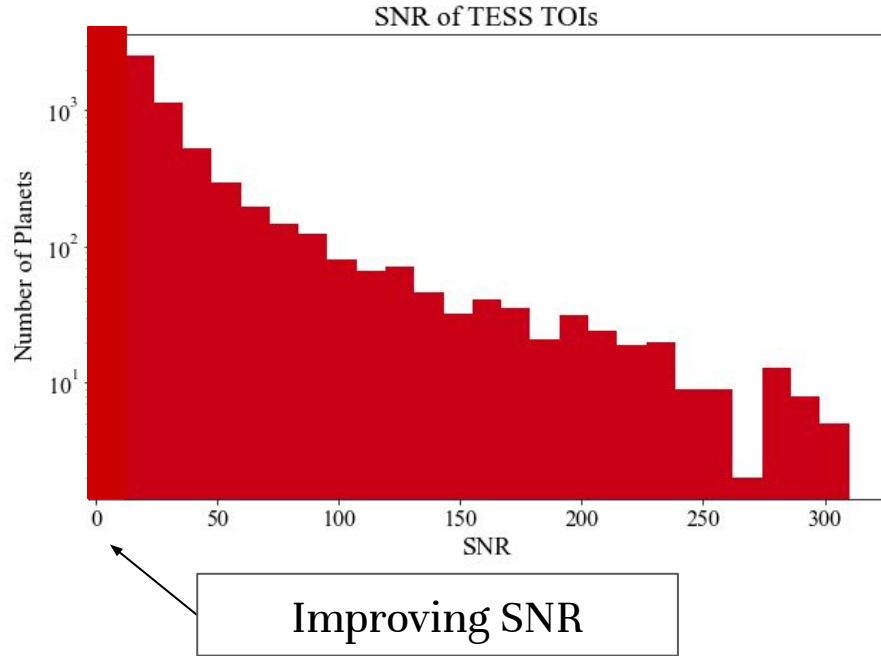
I propose using a **nonlinear neural network (an Autoencoder)** to similarly subtract systematic noise from the TESS light curves, with the aim of **detecting fainter planets and improving the measurement precision of existing planets.**

Motivations



- Most TESS Targets of Interest (TOIs) have low signal-to-noise ratios
- Improving noise correction can access this regime of noisier data
 - More Earth-like planets (smaller radii, longer periods), planets with active stellar hosts, etc.

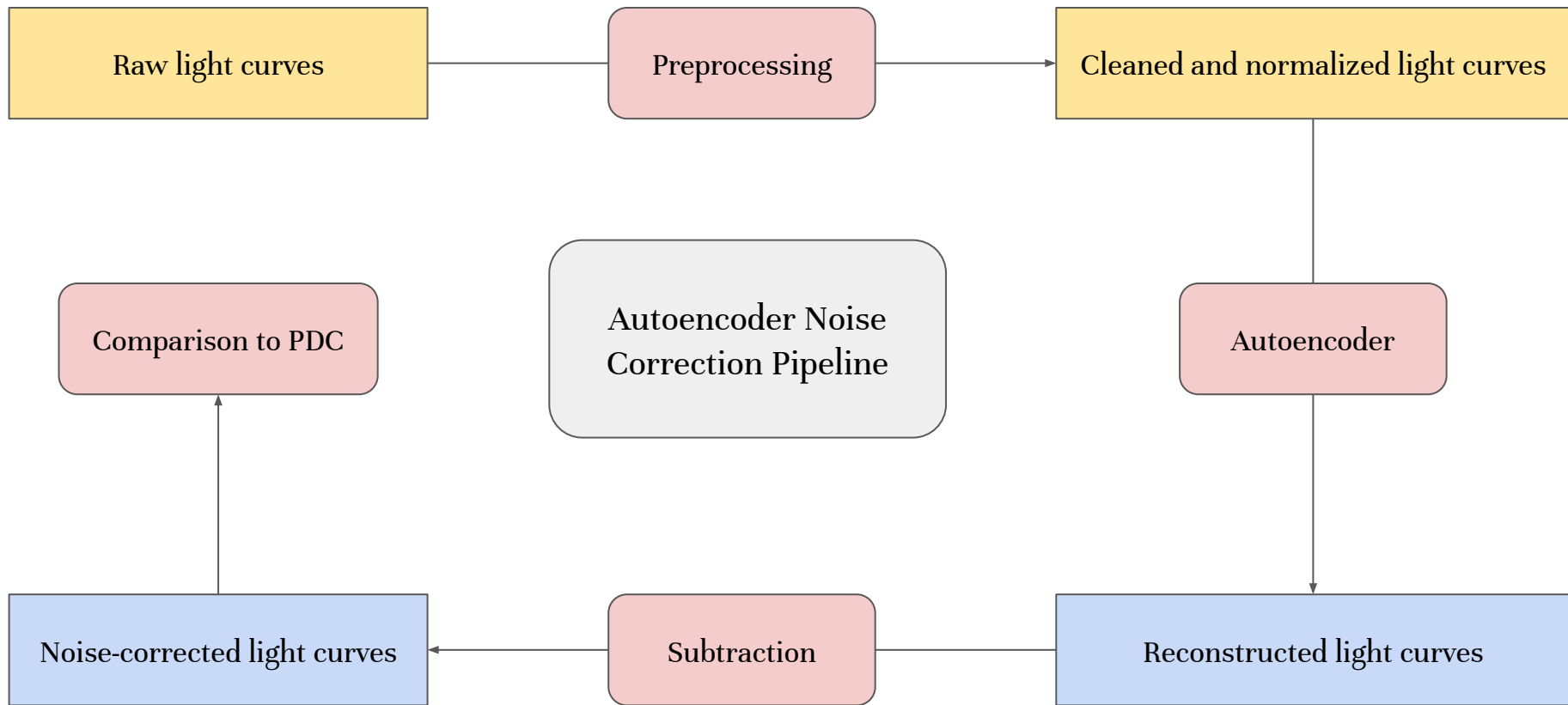
Motivations



- Most TESS Targets of Interest (TOIs) have low signal-to-noise ratios
- Improving noise correction can access this regime of noisier data
 - More Earth-like planets (smaller radii, longer periods), planets with active hosts, etc.
- Improving SNR by 1-2% will add hundreds of planets

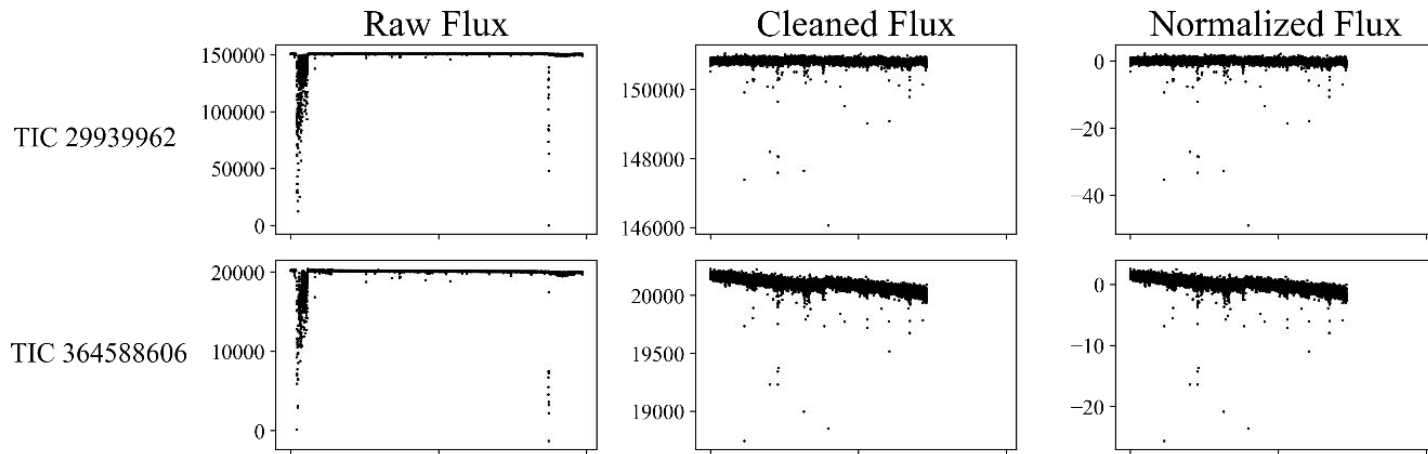
Why an Autoencoder?

- PDC is linear $\rightarrow$ Autoencoders are nonlinear neural networks
 - Nonlinearity allows us to model more complex relationships and correlations in data
- Applying a nonlinear algorithm could potentially improve the noise subtraction pipeline
- Autoencoder could also be tuned to remove stellar variability as well (don't need to be as conservative)

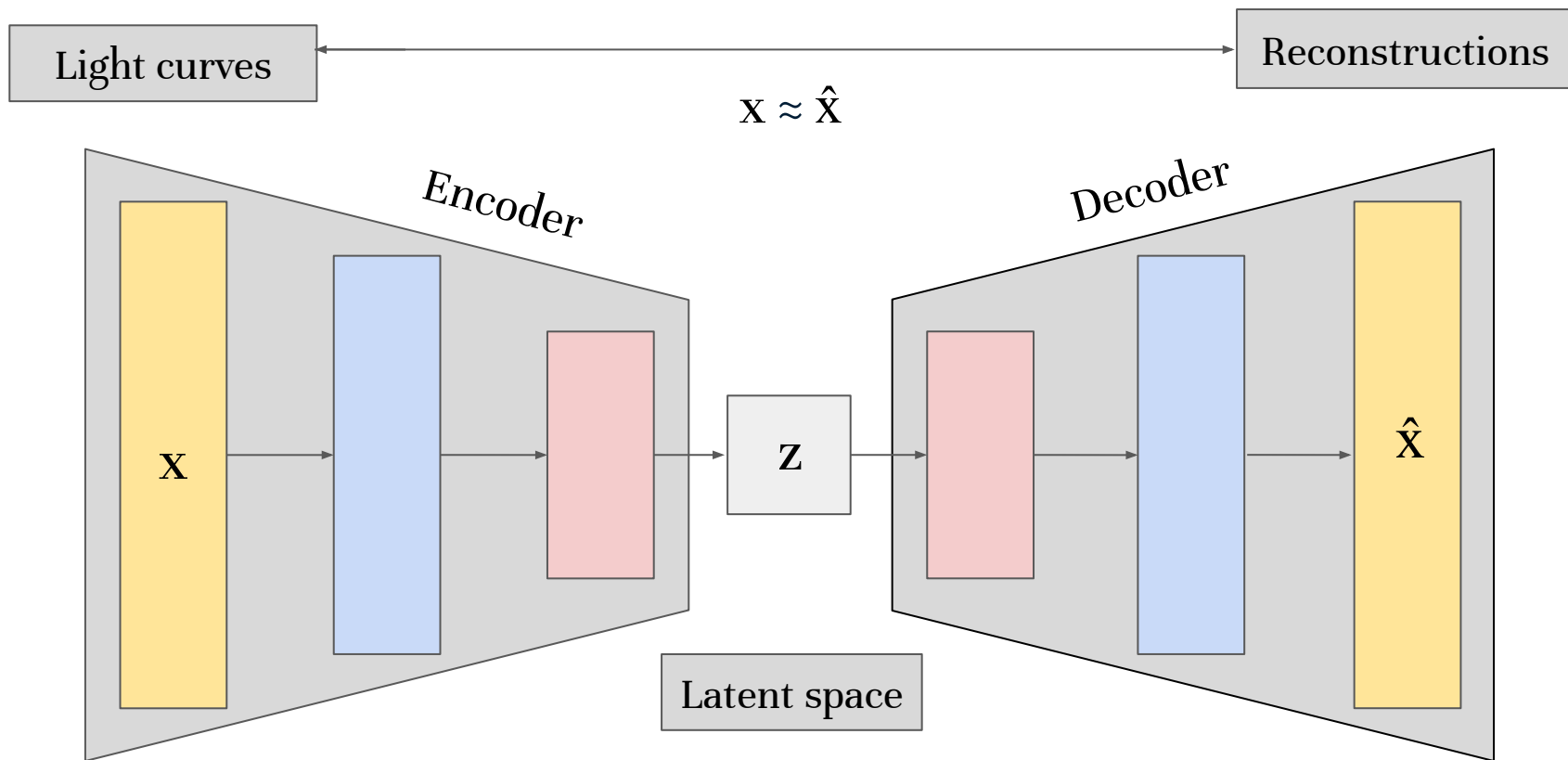


Preprocessing Data (Sector 3)

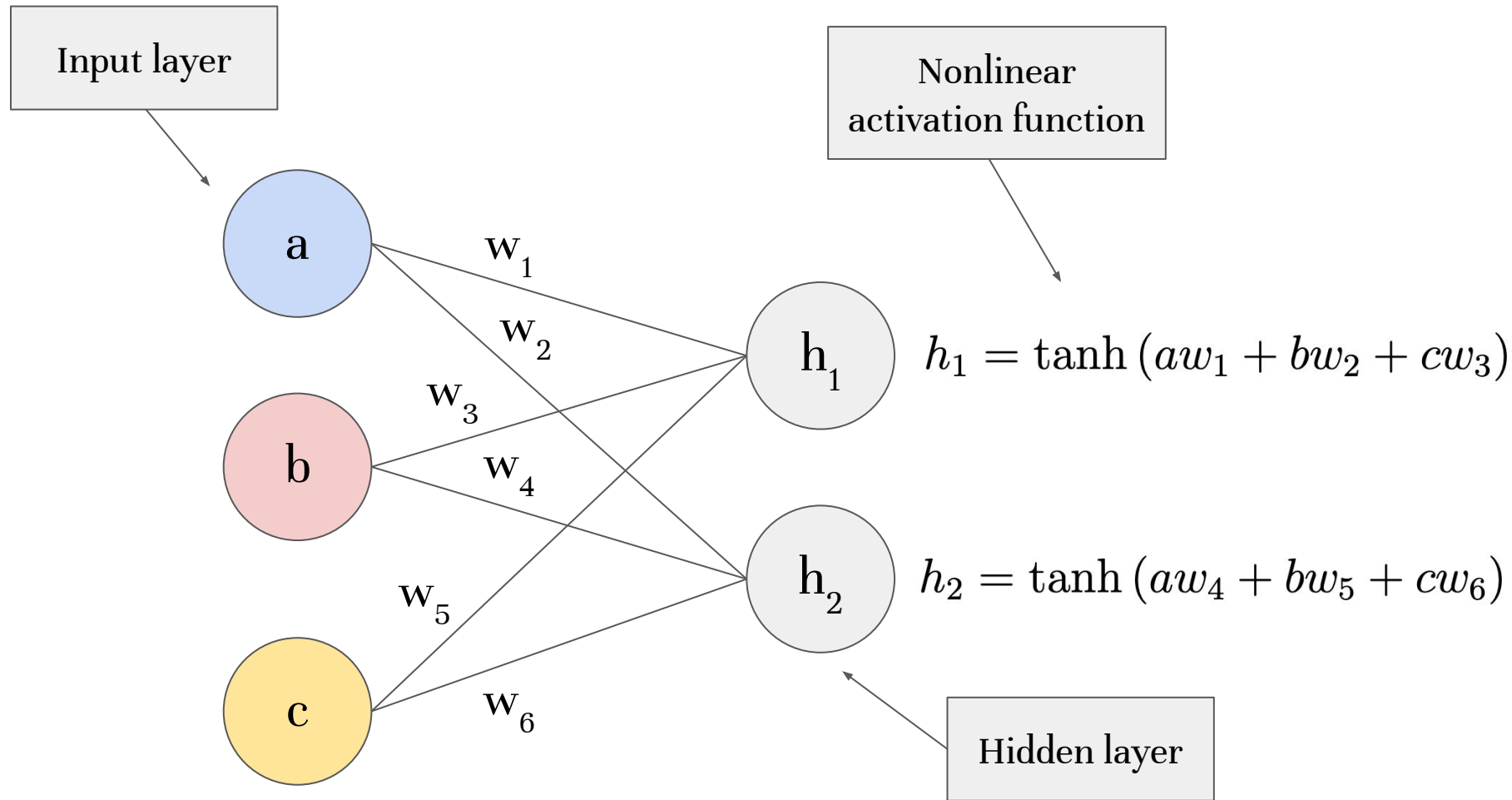
- Need to preprocess and normalize light curves first to remove excess noise and outliers
- Trimmed and removed points that TESS flags as “bad quality”



Autoencoder Basics



raw light curves – reconstructed noise = better data!



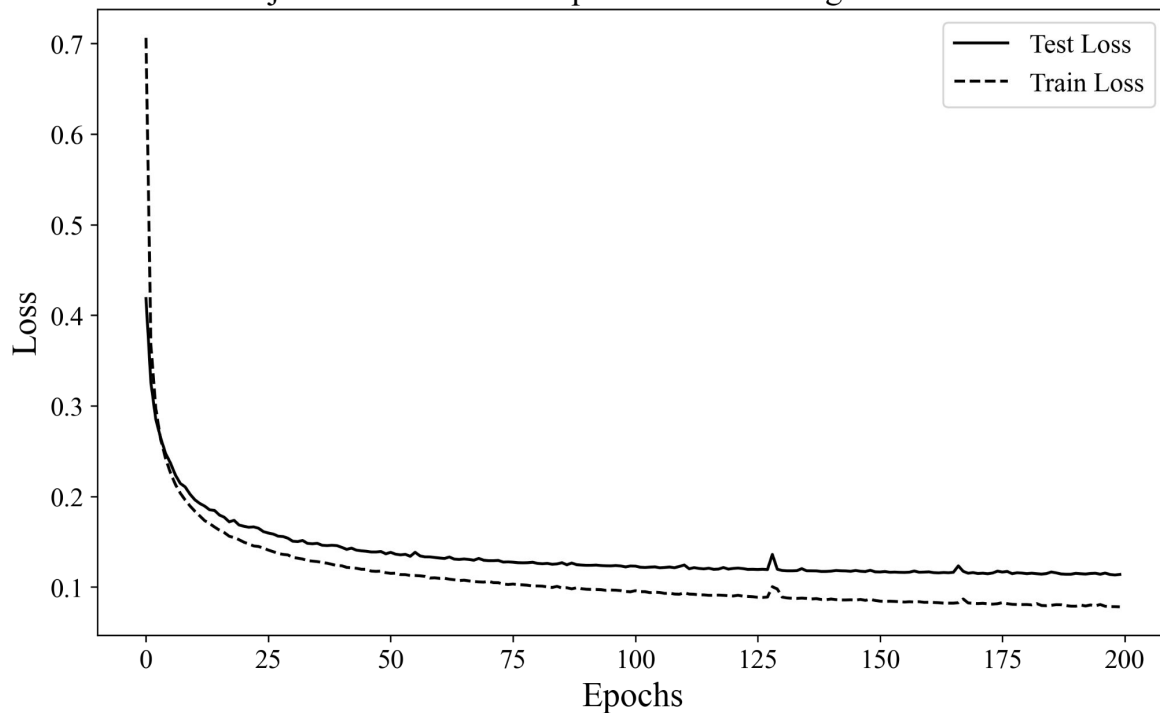
Autoencoder Training

- Trains by setting initial weights → calculating loss (reduced chi-squared) → adjusting weights to minimize chi-squared
- Aim to minimize loss and reconstruct light curve noise
- Can tune size of neural network and learning rate

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (x_i - \hat{x}_i)^2$$

Autoencoder Training

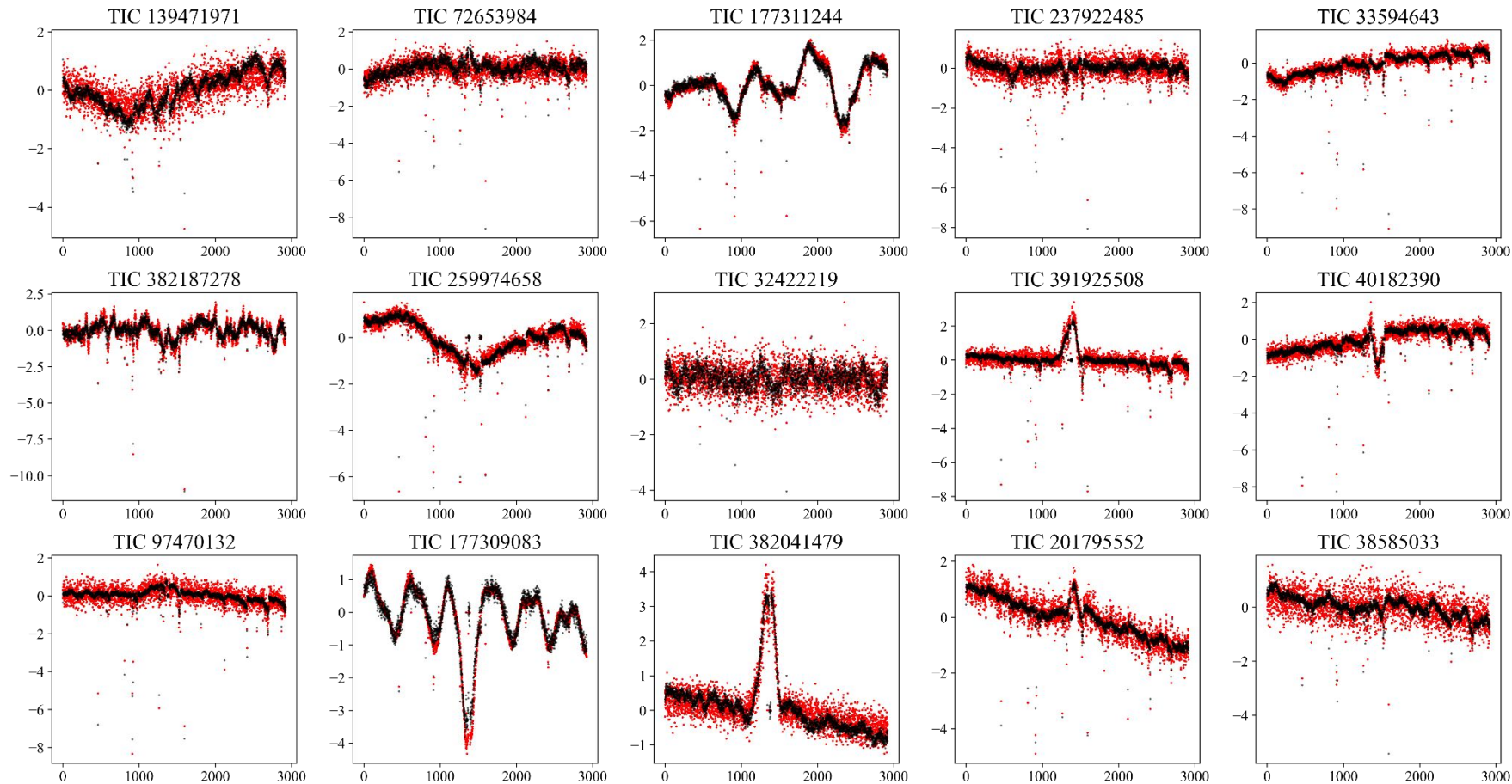
Objective Function vs. Epoch with Learning Rate of 0.472



Parameter	Value
Input Size	2920
Number of Layers	2
First Layer Size	512
Second Layer Size	256
Batch Size	128
Learning Rate	0.4722
Momentum	0.9
Epochs	200
Final Train Loss	0.1158
Final Test Loss	0.1377

Red = original
Black = reconstruction

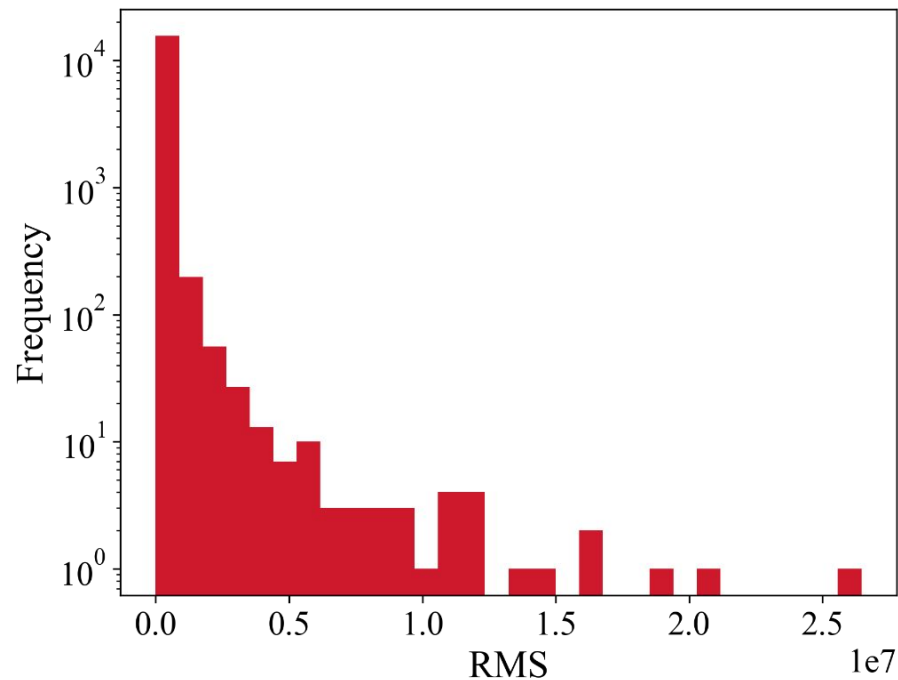
Autoencoder Reconstructions



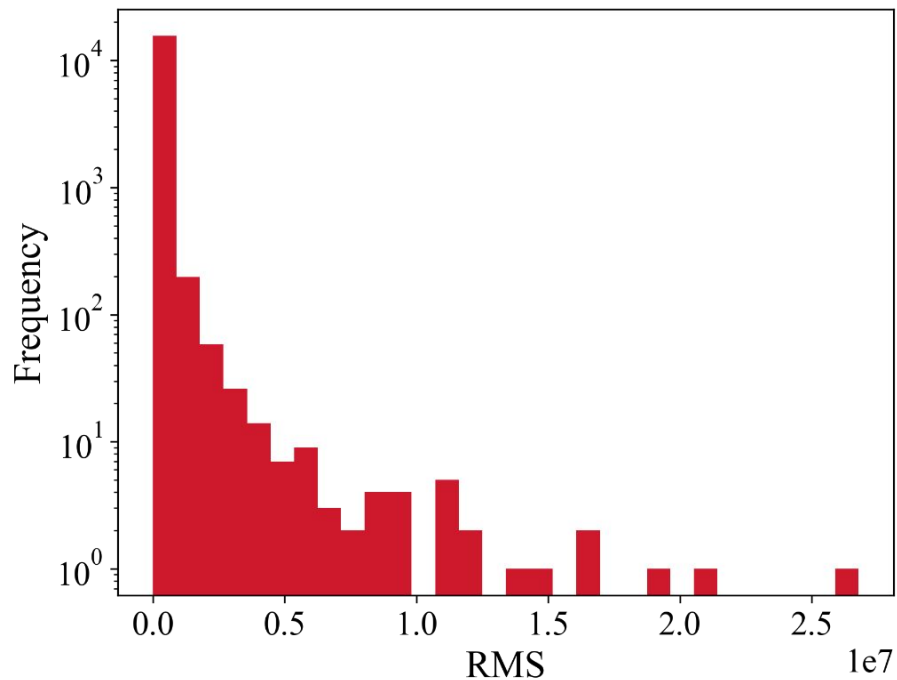
Autoencoder is
0.3% “better”

Comparing RMS

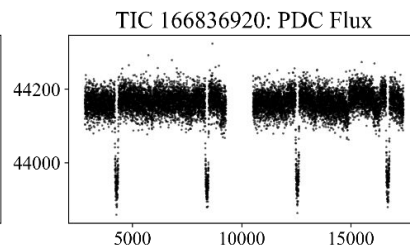
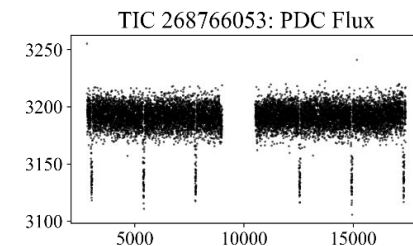
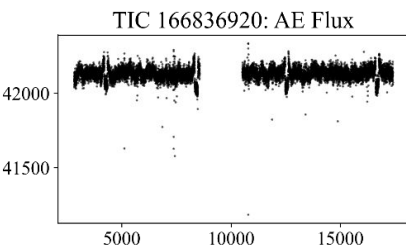
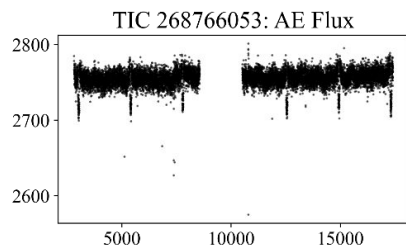
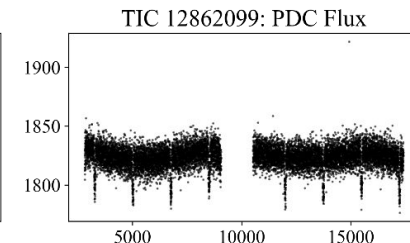
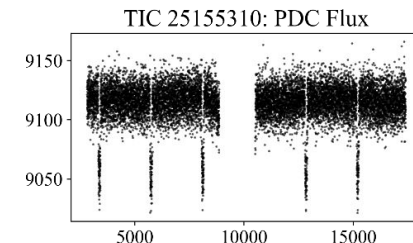
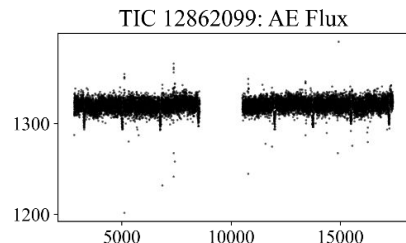
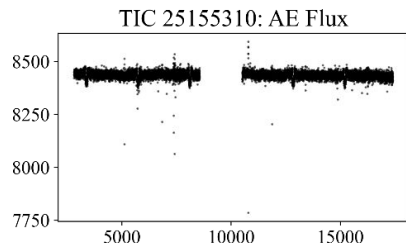
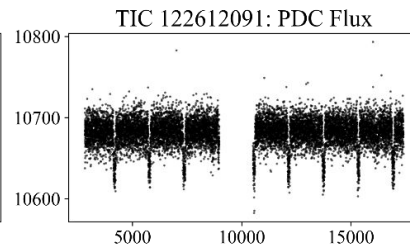
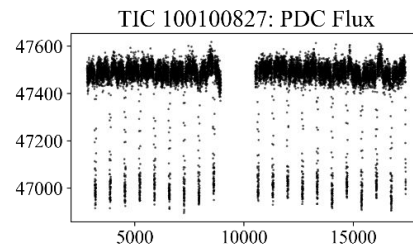
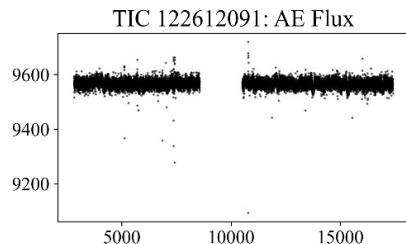
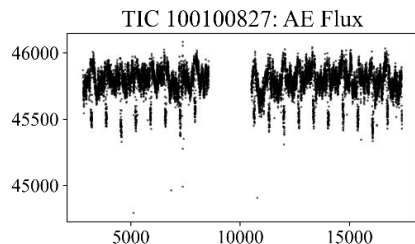
RMS Distribution: Autoencoder



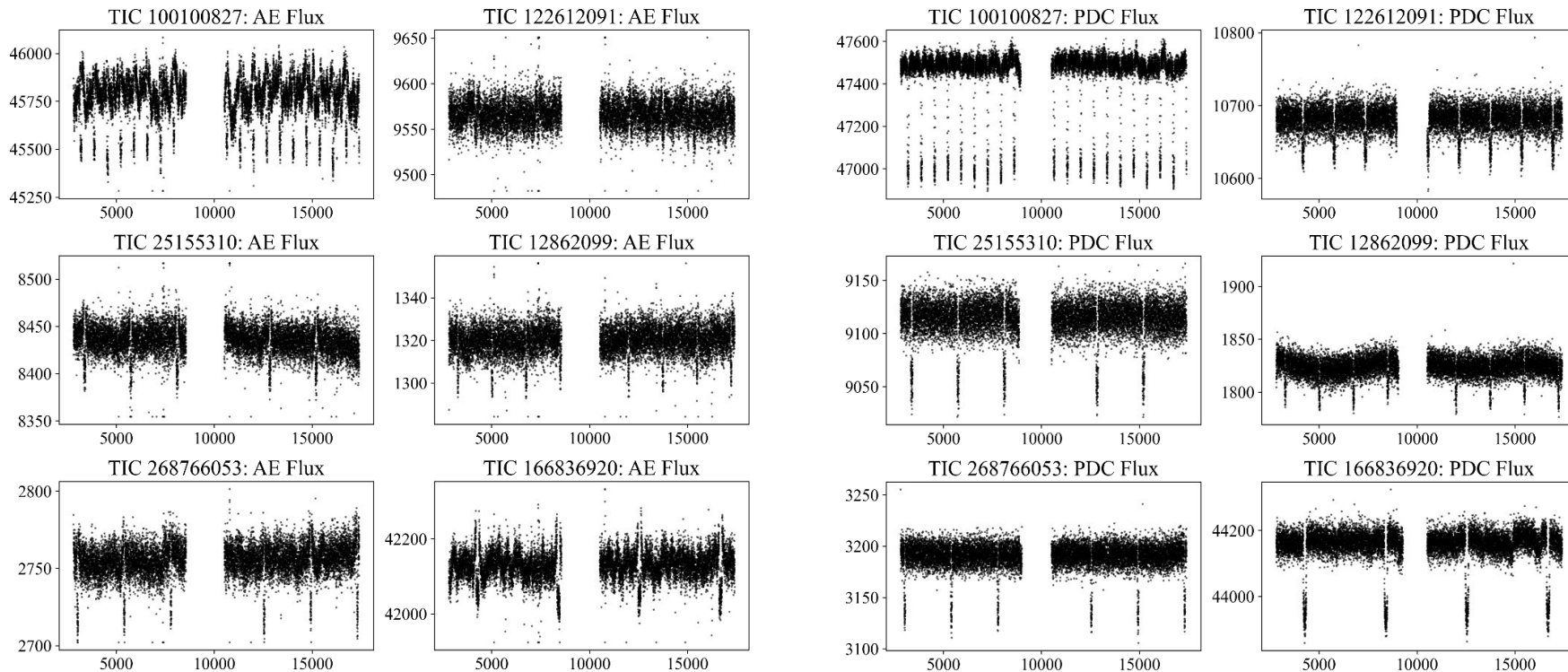
RMS Distribution: PDC



Comparing Known Planets

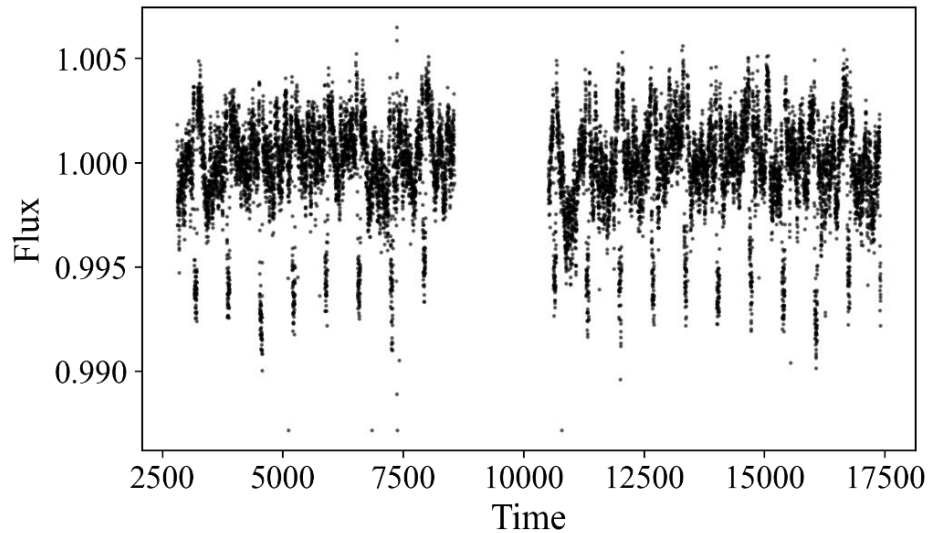


Comparing Confirmed Planets (with Sigma-Clipping)

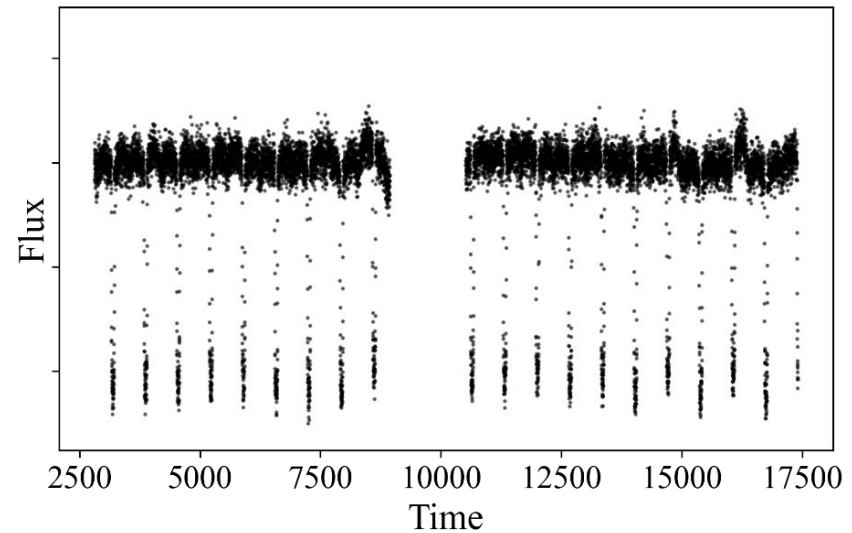


Comparing Planets: WASP-18 b

AE flux for TIC 100100827



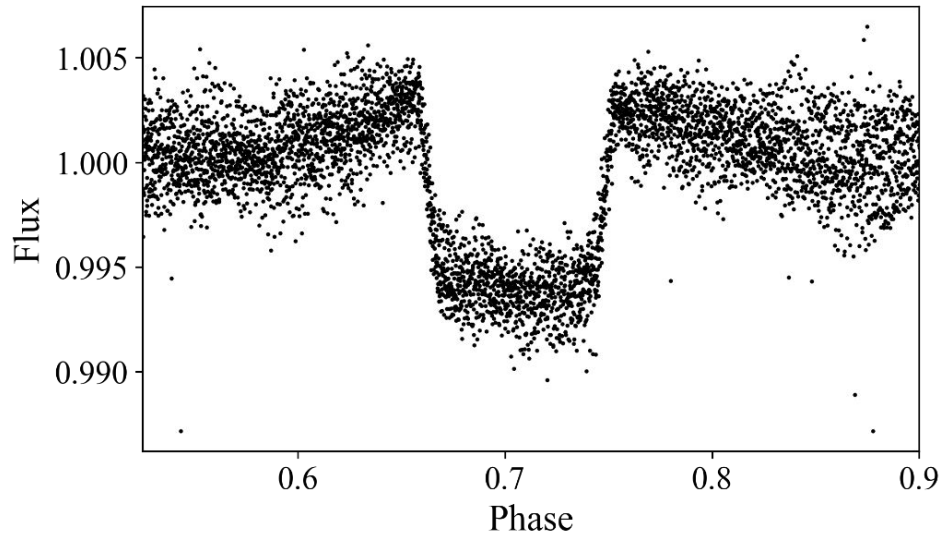
PDC flux for TIC 100100827



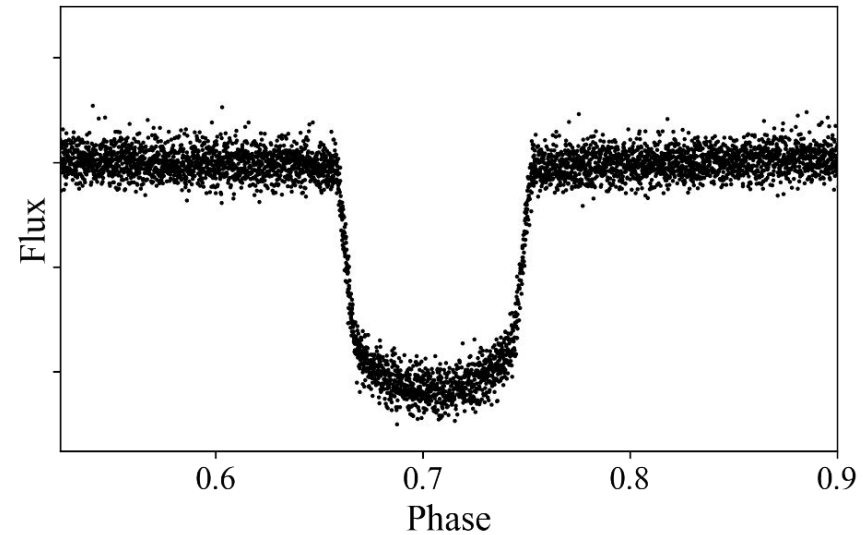
Hot Jupiter with orbital period of 0.9 days

Comparing Planets: WASP-18 b

AE flux for TIC 100100827



PDC flux for TIC 100100827

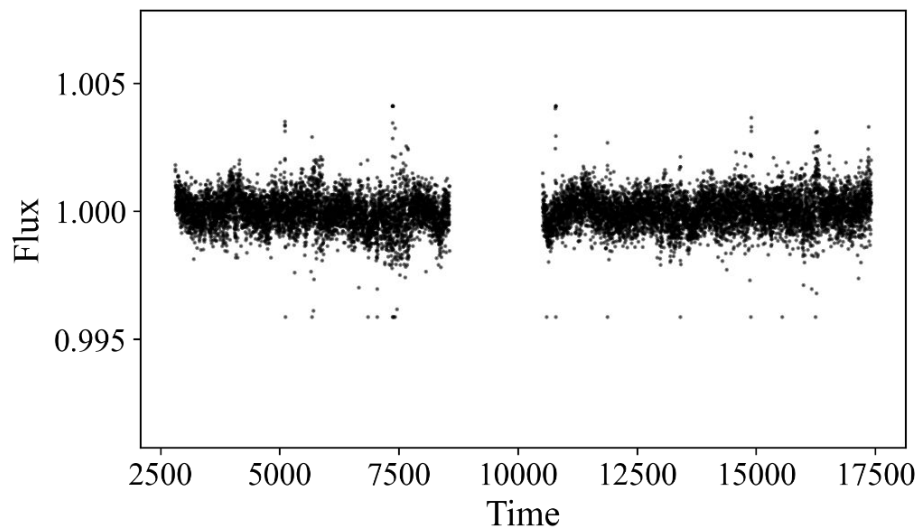


Hot Jupiter with orbital period of 0.9 days

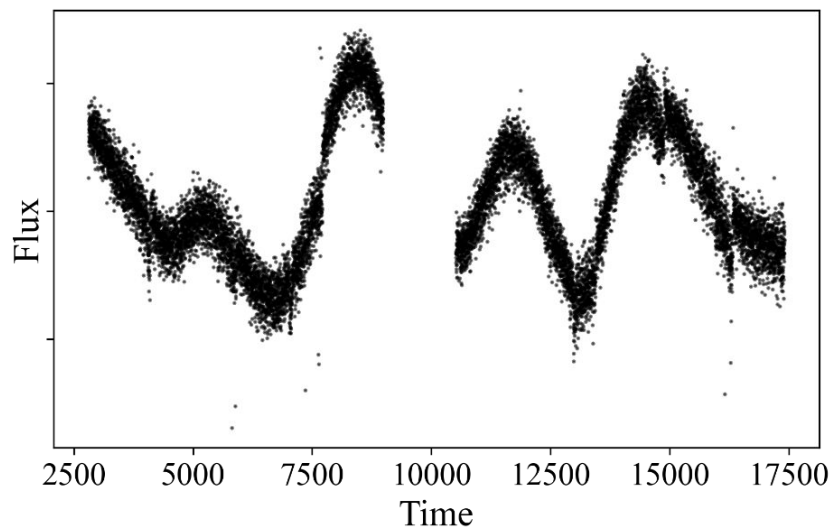
Comparing Planets: HD 18599 b

Confirmed by TESS!

AE flux for TIC 207141131



PDC flux for TIC 207141131

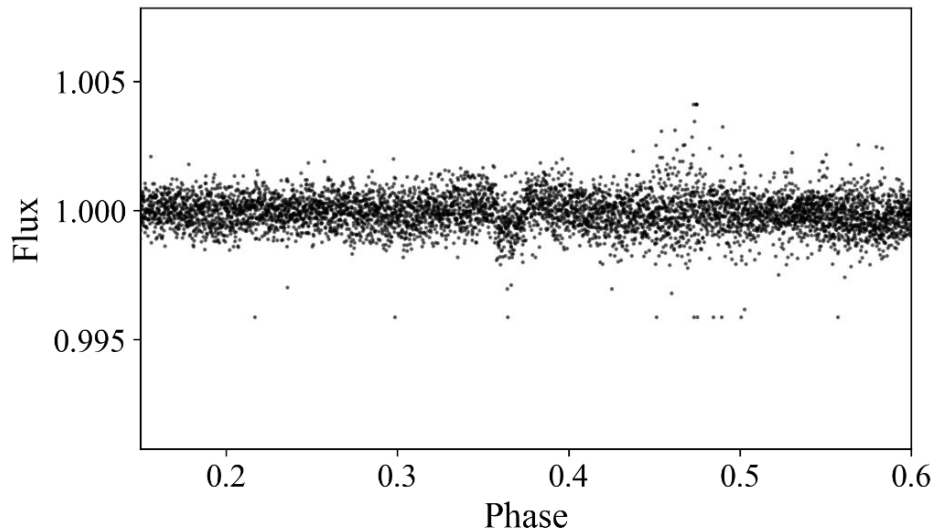


Young Neptune-like planet with orbital period of 4.1 days and active host star

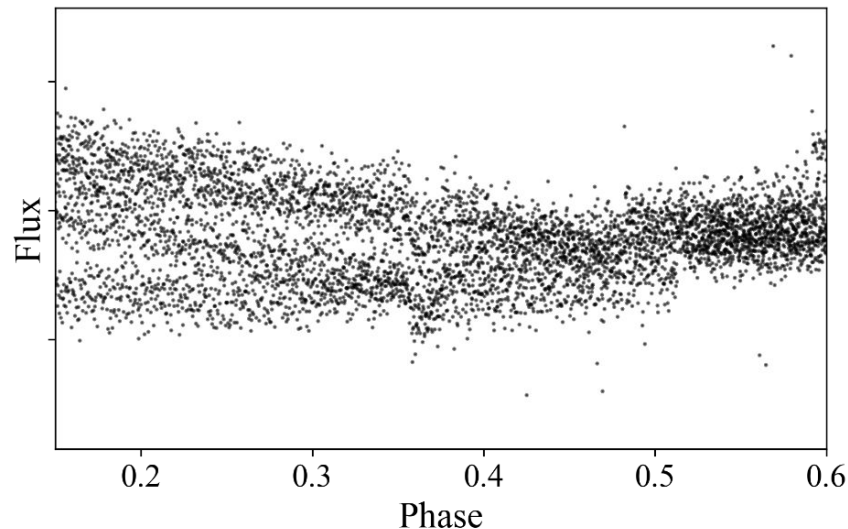
Comparing Planets: HD 18599 b

Confirmed by TESS!

AE flux for TIC 207141131



PDC flux for TIC 207141131



Young Neptune-like planet with orbital period of 4.1 days and active host star

Future Steps

- Need to preserve transit signal!
 - Explore preprocessing techniques (sigma-clipping, masking transits, binning, changing time scales, outlier rejection, etc.)
 - Test different neural network parameters (tuning layer size, number of layers, learning rate)
- Run Box Least Squares periodogram to look for more transit signals
 - If we improve RMS by even 1-2%, we access a whole new regime of planets with low SNR
- Train subsequent sectors

Acknowledgements

Thank you to my senior thesis advisor Dr. Jason Eastman as well as Dr. Chris Shallue and Professor Andrew Vanderburg for their advice and guidance!

Thank you to Professor Hernquist and all of my classmates in Astro 99 for their support throughout the thesis!



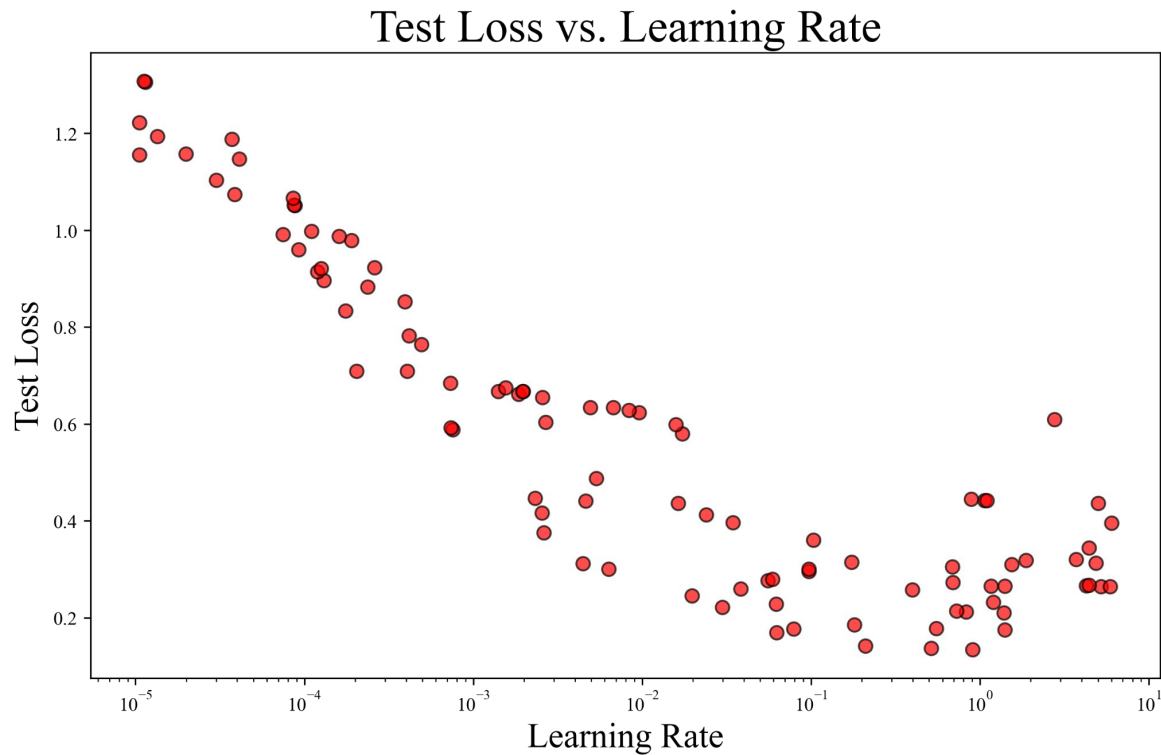
HARVARD
UNIVERSITY

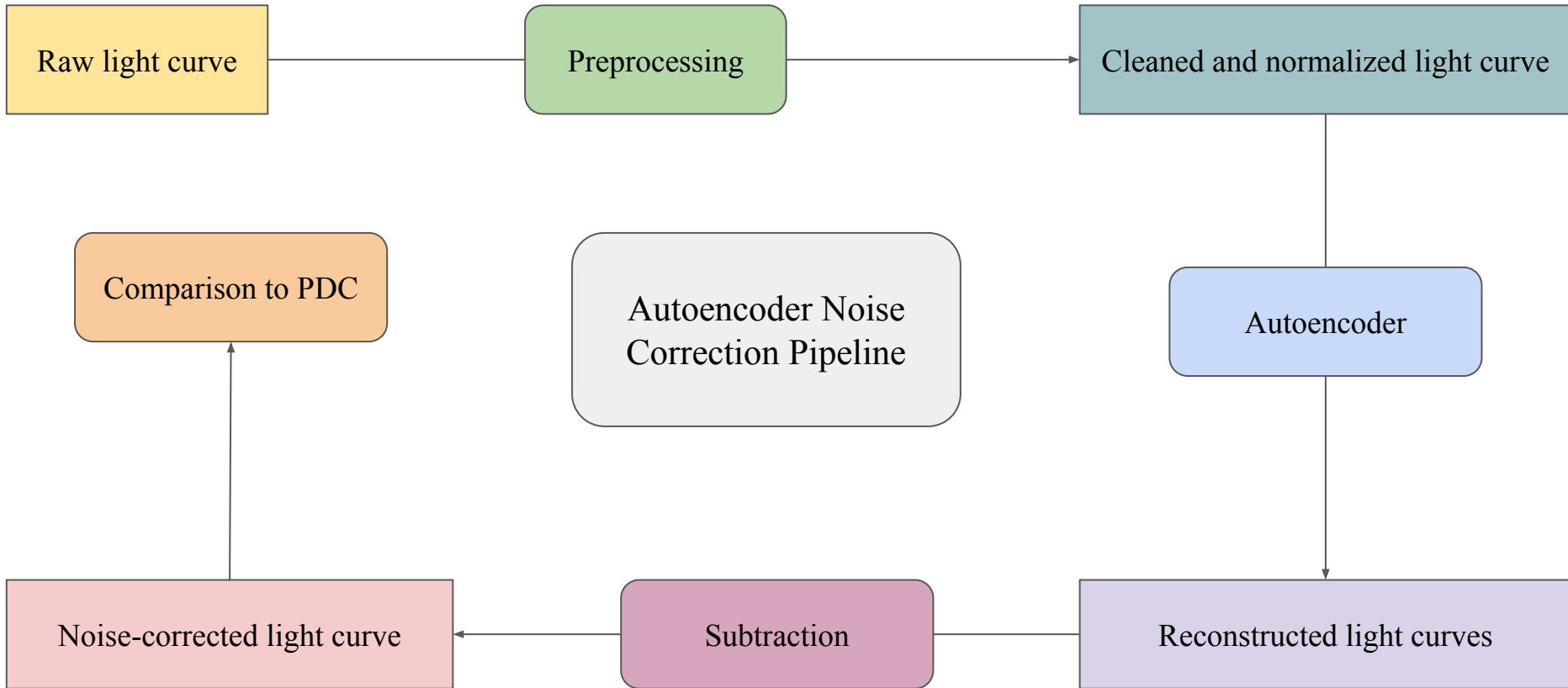
CENTER FOR

ASTROPHYSICS

HARVARD & SMITHSONIAN

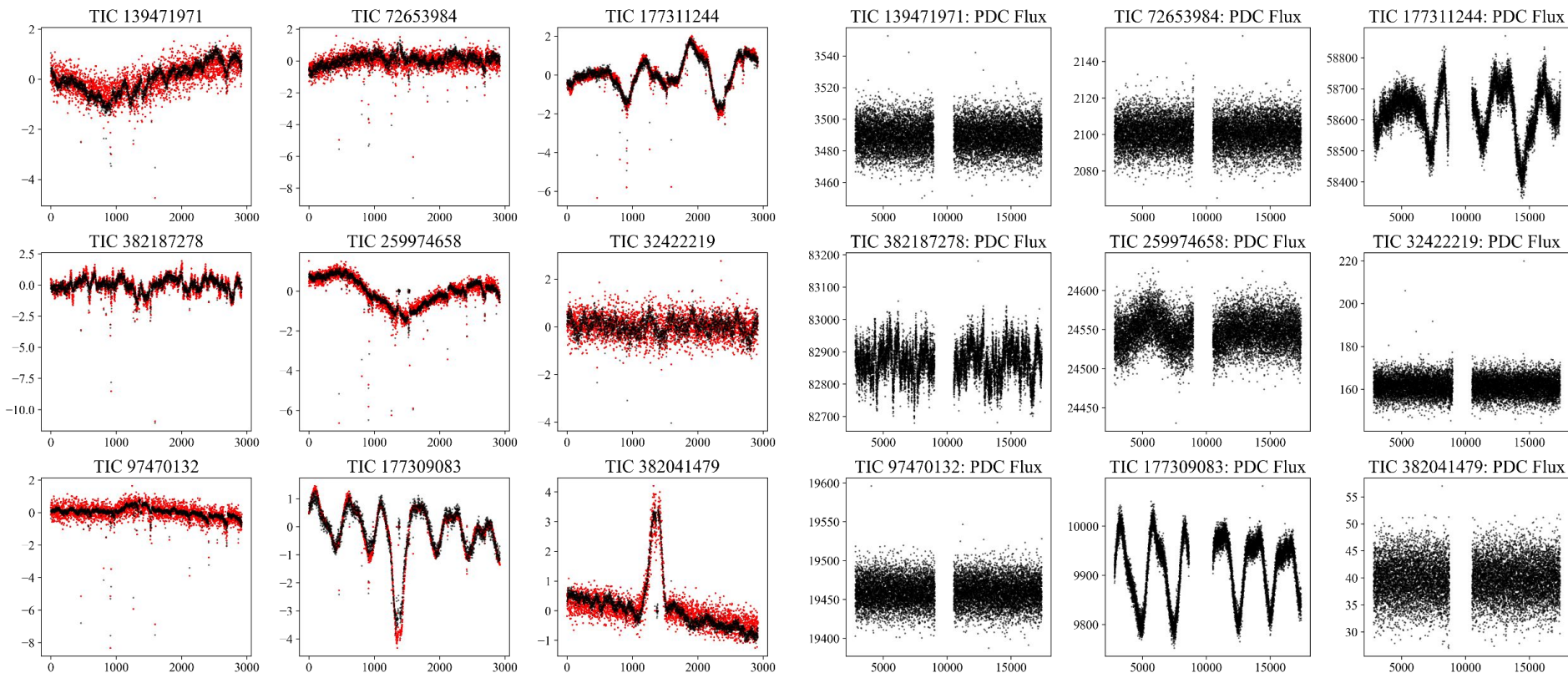
Random Search

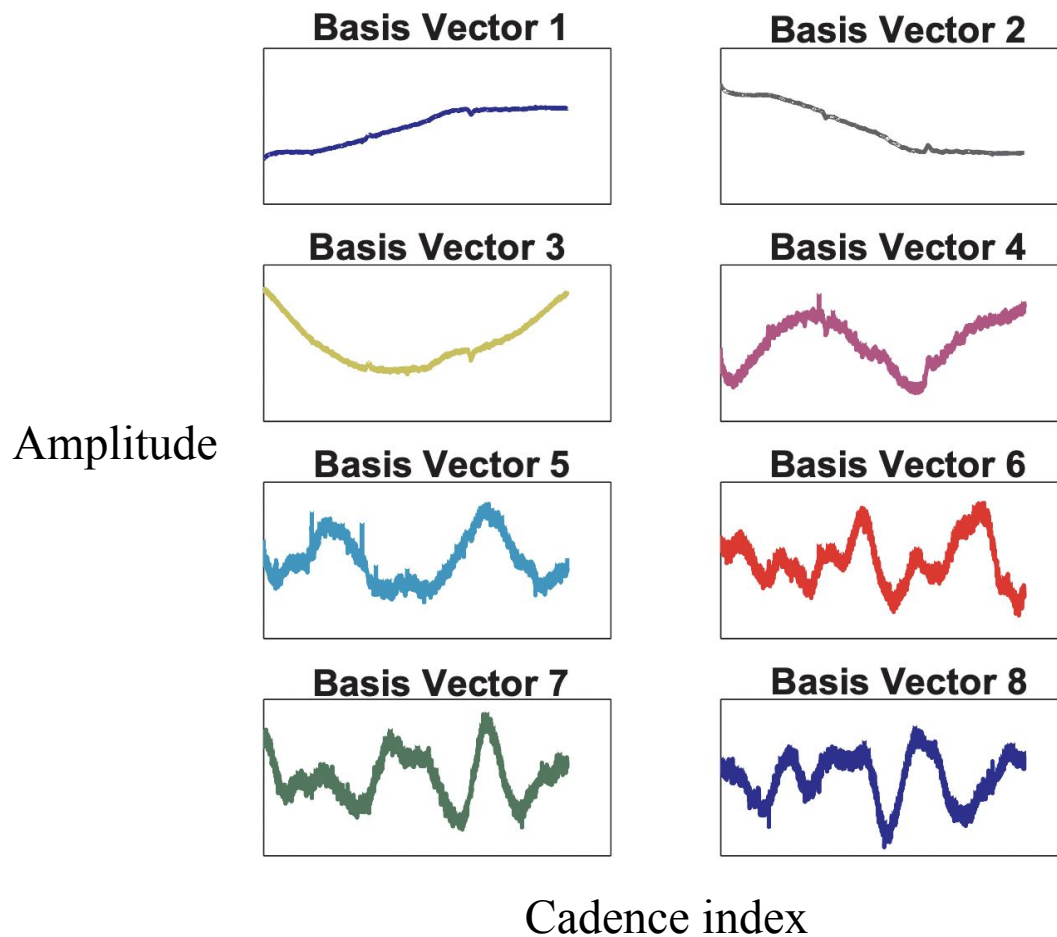




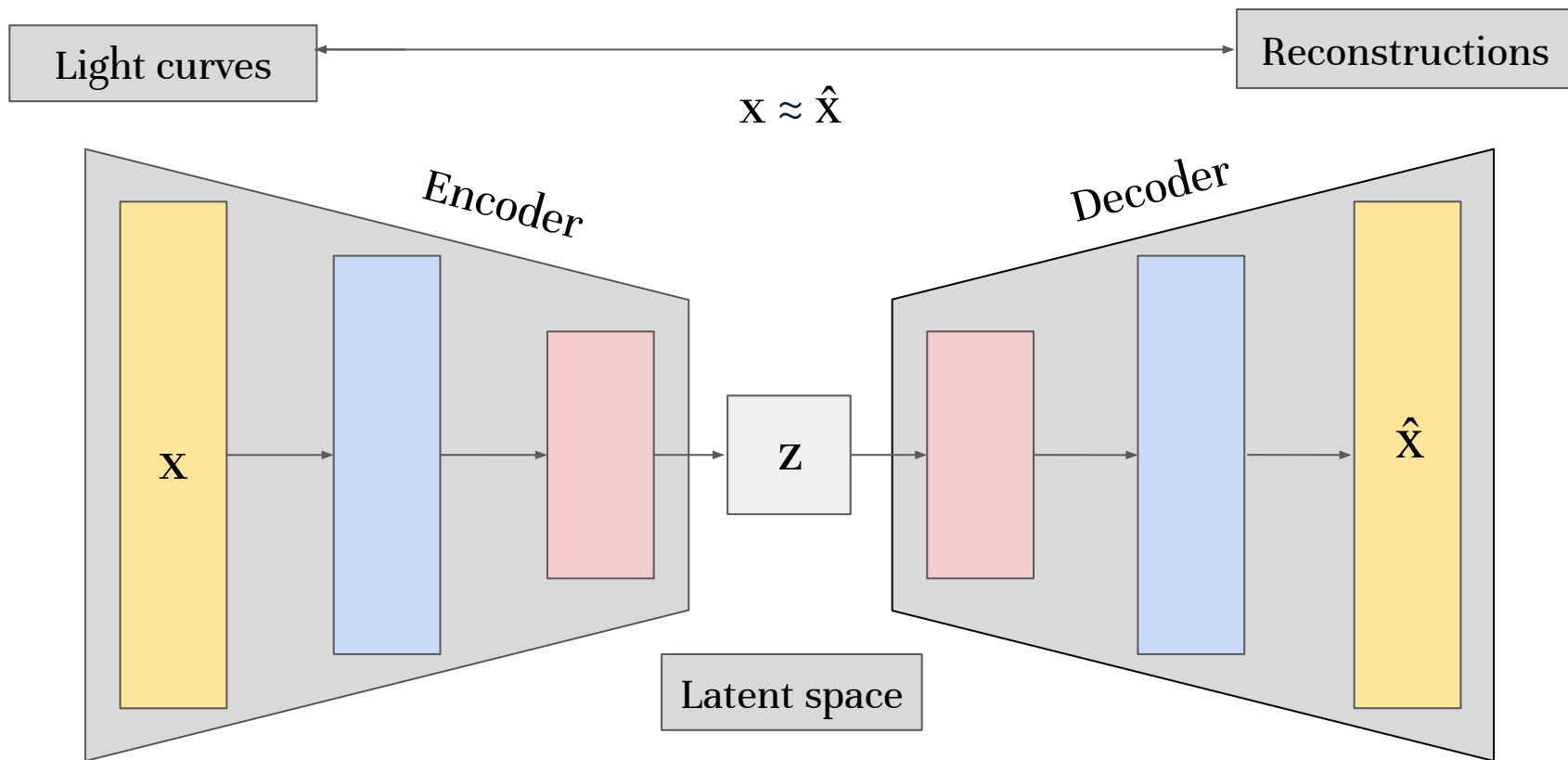
Red = original
Black = reconstruction

Autoencoder Reconstructions





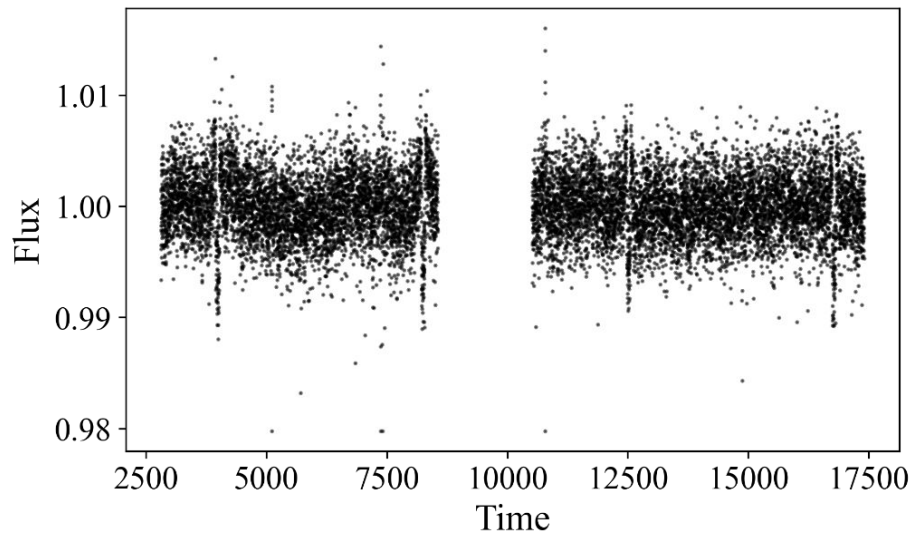
Autoencoder Basics



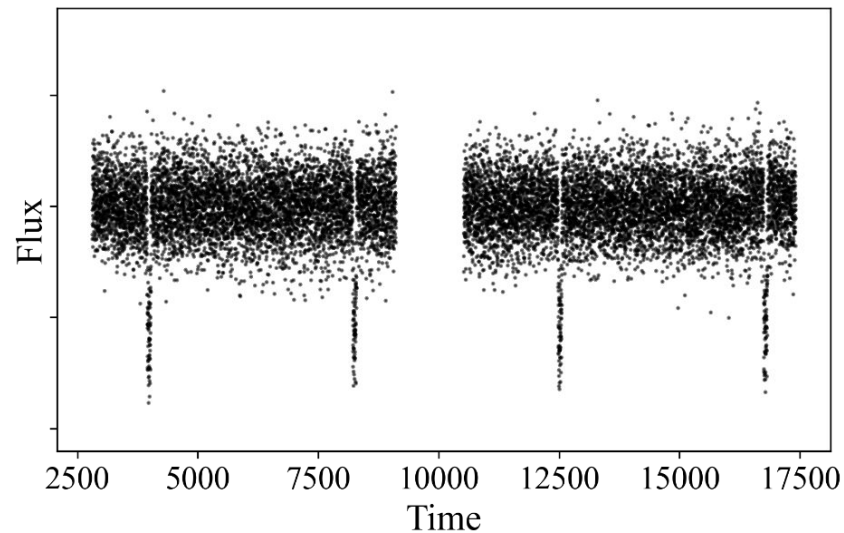
raw light curves – reconstructed noise = better data!

Comparing Planets: WASP-139 b

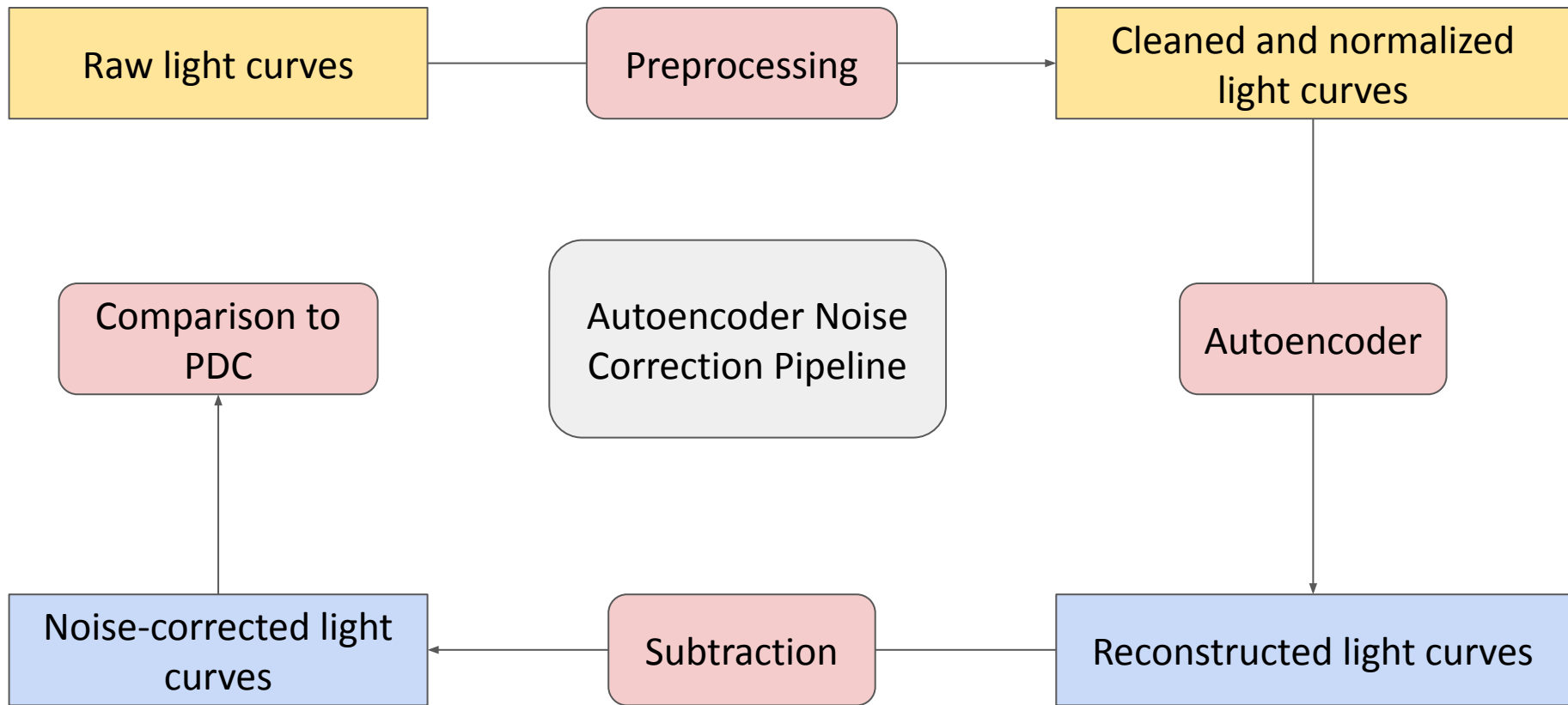
AE flux for TIC 159951311

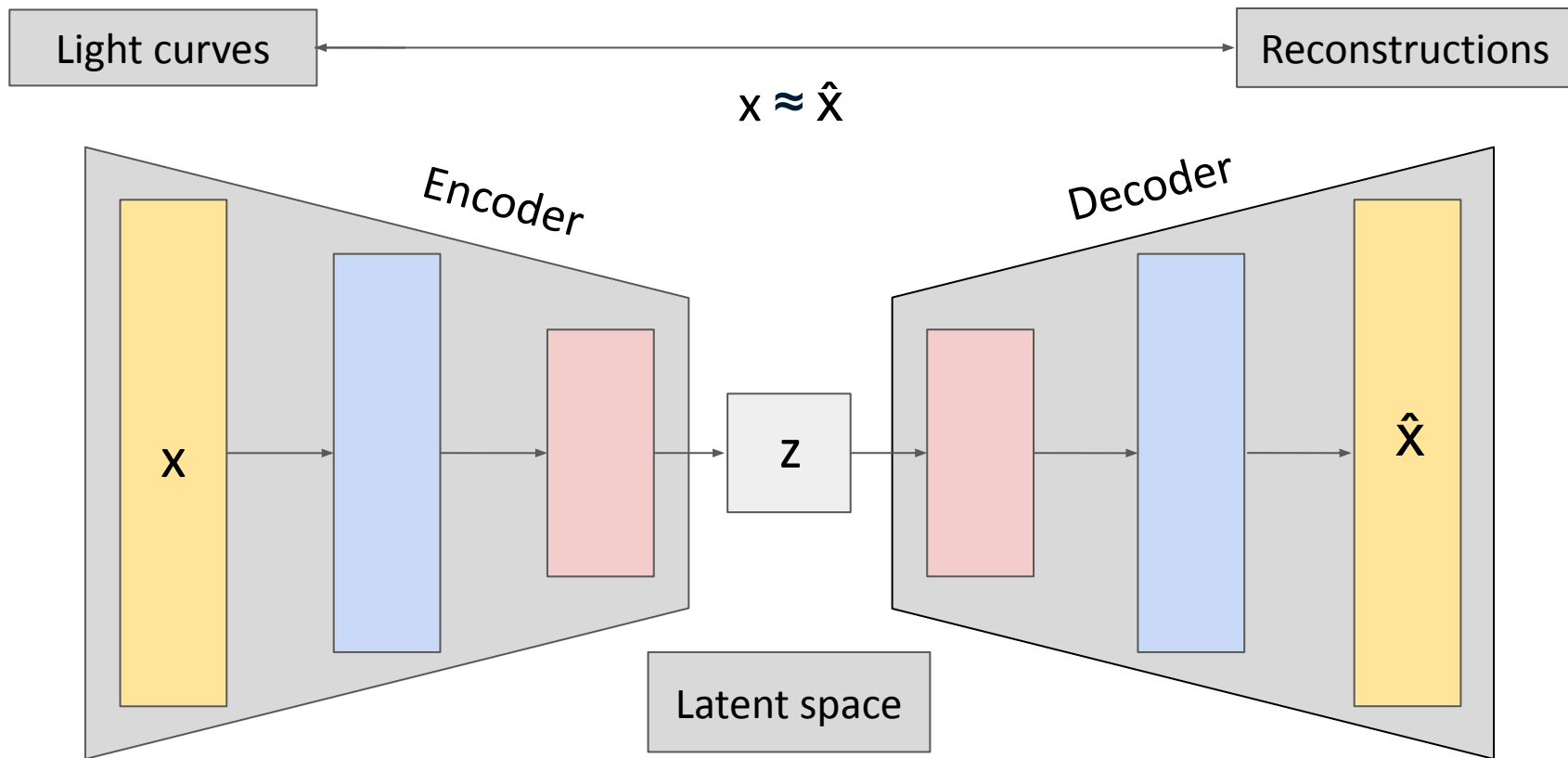


PDC flux for TIC 159951311



Super-Neptune with orbital period of 5.9 days

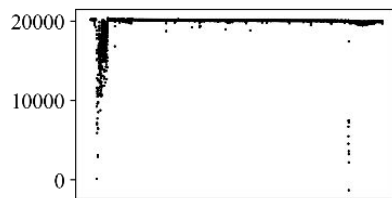




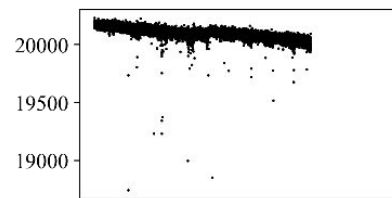
raw light curves – reconstructed noise = better data!

TIC 364588606

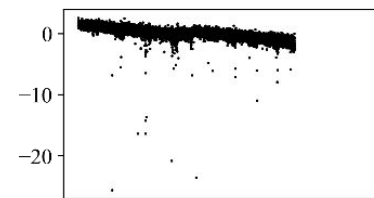
Raw flux



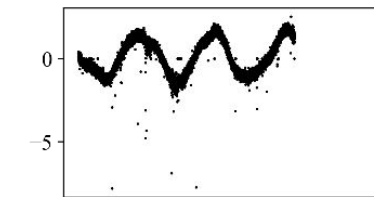
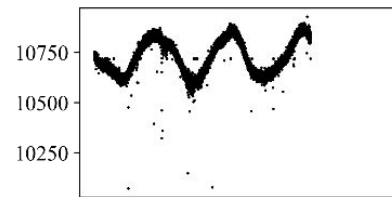
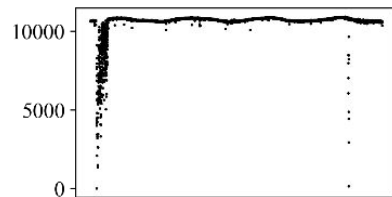
Cleaned flux



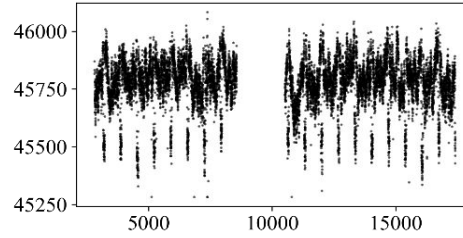
Normalized flux



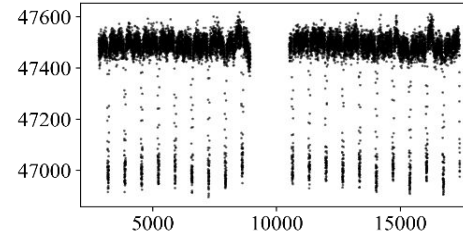
TIC 167164775



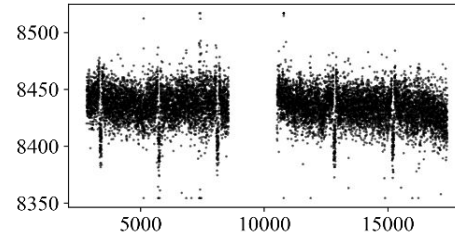
TIC 100100827: AE Flux



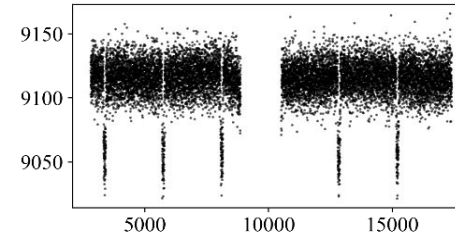
TIC 100100827: PDC Flux



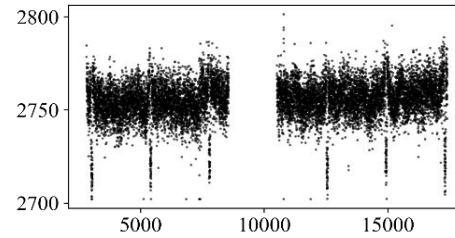
TIC 25155310: AE Flux



TIC 25155310: PDC Flux



TIC 268766053: AE Flux



TIC 268766053: PDC Flux

